Steve Berggreen

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Research Interests

Global environmental change; climate impacts; hydrology and water resources; human health and development.
Methods: hydrological modeling, geospatial analysis, remote sensing, causal inference, and machine learning.

Academic Appointments

Stanford University January 2025–present

Postdoctoral Scholar, Global Policy Laboratory, Stanford Doerr School of Sustainability
Principal investigator: Solomon Hsiang

Education

Lund University 2024

Ph.D. in Economics
Supervisors: Therese Nilsson and Gunes Gokmen

University of Gothenburg 2016

M.Sc. in Economics, with distinction

Lund University

M.Sc. in Water Resources Engineering 2011

B.Sc. in Civil Engineering 2009

Research visit: London School of Economics, Visiting Research Student, 2023.
Exchange study: ENSEEIHT, Hydraulics and Fluid Mechanics, Erasmus scholarship, 2010.

Professional Experience in Hydrology and Climate Science

Danish Hydraulic Institute 2016–2019

Development and application of methods for modeling urban flash floods.

Swedish Meteorological and Hydrological Institute 2014–2016

Contributed to applied hydrological and climate-impact modeling for national and regional climate-adaptation assessments, including downscaled climate scenarios, future streamflow, and flood risk.

WSP Group 2011–2014

Hydrological consulting and project management in flood risk and water-sensitive planning.

Peer-Reviewed Publications

A Robot-Assisted Pipeline to Rapidly Scan 1.7 Million Historical Aerial Photographs

Sheila Masson, Alan Potts, Allan Williams, **Steve Berggreen**, Kevin McLaren, Sam Martin, Eugenio Noda, Nicklas Nordfors, Nic Ruecroft, Hannah Druckenmiller, Solomon Hsiang, Andreas Madestam, and Anna Tompsett.
npj Heritage Science 14, article 123 (2026).

Job Market Paper

The Child Health Burden of Unobserved Water

With Linn Mattisson.

Working Papers

Estimating the effect of soil moisture on farmer adaptation and famine in Africa

With Hannah Druckenmiller and Solomon Hsiang.

When the Right Signal Fails: Climate Risk, Precautionary Saving, and the Limits of Self-Insurance

Under review.

The Agricultural Origins of Time Preference: Comment

Revise and resubmit, American Economic Review.

Demand for Slaves, Markets, and Social Betrayal: How the Slave Trade Shaped Mistrust in Africa

With Gunes Gokmen.

Under review.

Traditional Norms and Parental Investment in Human Capital

With Gunes Gokmen.

Research in Progress

Climate Adaptation in Africa over the 20th Century

With Hannah Druckenmiller, Dylan Hogan, and Solomon Hsiang.

Lead author, Aerial History Project.

Surface Water and Malaria Prediction in Uganda

With Shakira Babirye, Gabby Barratt Heitmann, Jade Benjamin-Chung, Grant Dorsey, Paul Krezanoski, Henry Mawejje, Erin Mordecai, Joaniter Nankabirwa, Mercy Opiyo, and Eddie Thomsen.

Field data collection ongoing.

The Gravity of Water: Water Scarcity and the Economic Geography of Floodplains

Manuscript in preparation.

One Million High-Resolution Aerial Photographs Capture Africa's 20th-Century Transformation

With Eugenio Noda, Hannah Druckenmiller, Joel Ferguson, Simon Greenhill, Solomon Hsiang, Taylor Kee, Andreas Madestam, Hikari Murayama, Nicklas Nordfors, Stephan Schneider, and Anna Tompsett.

Manuscript in preparation, Aerial History Project.

Cuisine

With Girija Borker and Margarita Gafaro.

Manuscript in preparation.

Policy Publications and Proceedings

Stagnant Water and Waterborne Diseases: Implications for Children's Health and Education

EBA Expert Group for Aid Studies, Development Dissertation Brief, 2025.

Traditionella Normer och Föräldrars Investering i Humankapital

With Jan Bietenbeck and Gunes Gokmen.

Ekonomisk Debatt, 2023; papers and proceedings of the 2022 Swedish National Conference in Economics.

Teaching and Mentoring

Research mentoring and supervision

2026– Undergraduate research mentor, Stanford University (ongoing).
2026 Undergraduate research mentor, Stanford SURGE (Khussam Mohamed).
2019 M.Sc. thesis supervisor for Lisa Olsson, Water Resources Engineering.

Teaching assistantships

2020–2024 Financial Economics (multiple course offerings).
2021–2023 Foundations of Finance.
2020–2022 Microeconomic Theory with Applications.
2019–2020 Macroeconomic Theory and Economic Policy.
2011 Water and Wastewater Technology.

Fellowships, Grants, and Awards

2026 Stanford SURGE program funding, USD 5,000, supporting undergraduate research at Stanford.

2023 Conference grant, Stiftelsen för främjande av ekonomisk forskning, SEK 7,500.

2022 Hedelius Scholarship for postgraduate studies at foreign universities, SEK 300,000.

2022 Paper selected for the Swedish National Conference in Economics papers and proceedings, *Ekonomisk Debatt*.

2016 Malmsten Award for Best M.Sc. Thesis in Economics, SEK 10,000. Thesis: *Flood risk and market property valuation when insurance is free*.

2009 Erasmus scholarship for studies at ENSEEIHT, Toulouse, SEK 40,000.

Invited Talks

2024 1st Development Economics Workshop, Durham University Business School.

2023 Development Economics Research Group, University of Copenhagen; IFS/STICERD (LSE)/UCL Development Work-in-Progress Seminar, IFS; Economics of the Energy and Environment Seminar, LSE; Copenhagen Business School; ENS Lyon.

2022 Young Leaders Programme Personal Finance Workshop, Friedrich Naumann Foundation.

Conference and Seminar Presentations

2026 NEUDC, Cornell University (upcoming); 6th Arne Ryde Workshop on Culture, Institutions and Development, Lund (upcoming); Midwest International Economic Development Conference, University of Wisconsin–Madison; Sustainability Data Science Conference, Stanford; GradSEEDS Seminar, Stanford.

2025 Environmental and Energy Economics Festival, Stanford; WASH Econ, Oberlin College; Global Environmental Policy Student Seminar, Stanford.

2024 8th Swedish Conference in Economics, Lund; PhD Conference on Sustainable Development, Lund; Development Tea, Stockholm University; Brown Bag Seminar, Lund (twice).

2023 Southern Economic Association Annual Meeting, New Orleans; CESifo Economics of Children Workshop, LMU Munich; NEUDC, Harvard; EALE, Prague; EEA, Barcelona; CSAE, Oxford; 3rd Arne Ryde Workshop on Culture, Institutions and Development, Lund; Essen Health Conference, University of Duisburg–Essen; Nordic Conference in Development, Gothenburg; EUDN PhD Workshop, Stockholm; DEM Workshop on Ageing, Culture and Comparative Development, University of Luxembourg; Economic Geography PhD Seminar, LSE; SWELL Seminar Series in Water Economics; Copenhagen Education Network Workshop, Lund.

2022 Swedish Conference in Economics, Stockholm School of Economics; IAST Workshop in Political Economy, Toulouse School of Economics; ASWEDE 7th Annual Conference, Stockholm; 2nd Arne Ryde Workshop on Culture, Institutions and Development, Lund; Brown Bag Seminar, Lund (twice); Toulouse Summer

School in Quantitative Social Sciences, Toulouse School of Economics; ASREC Graduate Student Workshop in the Economics of Religion, IRES, Chapman University.

2021 Arne Ryde Workshop on Culture, Institutions and Development, Lund; Copenhagen Education Network Workshop, Lund; Scandinavian PhD Workshop in Economics of Education.

Academic Service

Journal refereeing: Journal of Environmental Economics and Management; Economics & Human Biology.

2023–2024 Member, EDR doctoral student union election committee.
2023 Co-organizer, 3rd Arne Ryde Workshop on Culture, Institutions and Development; co-organizer, Economics PhD lunch seminar series, Lund University.
2022–2023 EDR doctoral student representative, Faculty Research Committee.
2022 Co-organizer, 2nd Arne Ryde Workshop on Culture, Institutions and Development; PhD representative, Head of Department election committee.
2021–2022 PhD representative, Department Board.
2019 Co-organizer, ASREC Europe Conference.

Technical Skills

Python, Stata, R; Google Earth Engine; ArcGIS, QGIS; LaTeX.

Languages

Swedish (fluent), English (fluent), French (advanced), Farsi (intermediate).

Academic References

Solomon Hsiang

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Hannah Druckenmiller

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Gunes Gokmen

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